

A Firm-Level Microsimulation for VAT Policy Analysis^{*}

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Preliminary and Incomplete. Comments are welcome.

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Abstract

I present an open, reproducible firm-level microsimulation for pricing reforms to the UK VAT registration threshold, released in full—including its synthetic data generator—as open-source code. The threshold is a *notch*: once turnover crosses £85,000, 20% VAT falls due on a firm’s entire turnover rather than only the excess, creating an incentive to bunch below. I build synthetic firm-level records from ONS business-structure counts (sector $\times$ turnover band $\times$ employment), calibrate them to HMRC VAT statistics for 2023–24, and apply three tools: a static costing of threshold-location reforms, a reduced-form bunching estimator, and a calibrated Kleven–Waseem notch model for schedule-shape reforms the reduced form cannot price. I treat the model as a turnover-tax-notch approximation for consumer-facing, limited-reclaim firms, abstracting from input reclaim and voluntary registration. Static costing reproduces HMRC’s published figure for the £85,000 to £90,000 reform (–£175m versus –£185m in 2025–26). Behavioural responses are small: bunching ratio $b \approx 0.061$; median local turnover elasticity ≈ 0.17 . I report each reform’s static cost and how much of the £21,250 dominated region it removes: raising the threshold to £100k (–£374m) leaves the £21,250 zone, relocated; a banded 10% reduced rate (–£357m) reduces it to £9,444; a 15% reduced rate (–£178m) reduces it to £15,000; and a graduated taper over [£85k, £105k] (–£335m) removes it.

Keywords: value-added tax, microsimulation, tax notch, policy

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1 Introduction

The United Kingdom’s Value Added Tax (VAT) applies above a registration threshold that is among the highest in the OECD (Seely, 2024; HM Revenue and Customs, 2024). The threshold is a *notch* rather than a kink: once a firm’s annual taxable turnover crosses it, the standard 20% rate falls due on all of the firm’s turnover, not only on the part above the threshold, so registration produces a discrete jump in liability rather than a marginal one. The threshold falls in a dense part of the firm distribution. Figure 1(a) shows that UK firms are concentrated at the lower end of the turnover distribution, and Figure 1(b) shows that VAT-registered firms are concentrated in the turnover bands at and below the threshold, so the notch applies to a large number of firms.

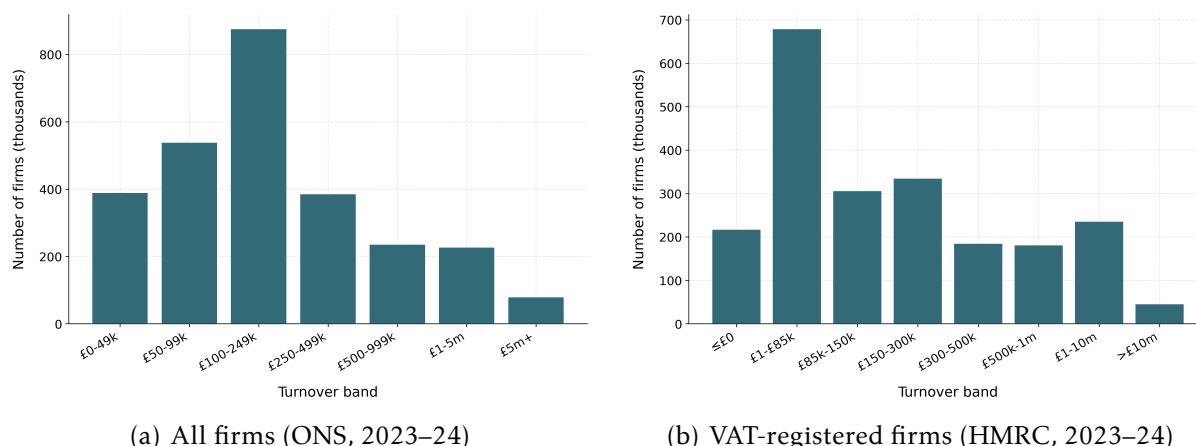


Figure 1: The UK firm-size distribution and VAT registration, 2023–24 (threshold £85,000). Panel (a): number of firms by turnover band. Panel (b): number of VAT-registered firms by turnover band. Source: panel (a), ONS, *UK Business: Activity, Size and Location* (firm counts by turnover band); panel (b), HMRC, *Value Added Tax (VAT) Annual Statistics* (VAT-registered population by turnover band).

Firms respond to the notch by locating below the threshold. Tabulations of HMRC turnover data show the number of firms rising towards £84,000 and then dropping sharply at the £85,000 threshold, leaving excess mass just below it (Office for Budget Responsibility, 2023). Liu et al. (2021) document this bunching in UK firm data, and Liu et al. (2024) find that firms slow their turnover growth as they approach the threshold. The notch therefore distorts the size distribution and the growth decisions of firms near the margin, which places the threshold under recurrent pressure for reform.

Reform proposals change the threshold in one of three ways: its *level* (raising or lowering it), its *shape* (replacing the notch with a schedule that phases the rate in over a band), or its *rate* (a reduced rate for firms in a band above the threshold). The three are not equivalent. Changing the level moves the notch along the turnover axis; changing the shape or the rate alters the size of the discrete jump itself. Costing the three on a

common basis is the practical difficulty this paper addresses. A change in the level can be costed by mechanical accounting, but a change in the shape of the schedule cannot be evaluated with the reduced-form bunching methods used to study the threshold, because a continuous schedule leaves no single point of excess mass to exploit. The firm-level data needed to study turnover near the threshold is, in addition, not available in open form.

In this paper I build an open, reproducible firm-level microsimulation for costing UK VAT threshold reforms. I construct synthetic firm microdata calibrated to published HMRC aggregates—band-level firm counts and sector turnover totals—for the 2023–24 tax year, when the threshold was £85,000 (raised to £90,000 in April 2024); the synthetic data can be released openly and regenerated under any counterfactual schedule. On these data I combine two methods. I measure the behavioural response with a reduced-form bunching estimator in the tradition of [Saez \(2010\)](#) and [Kleven and Waseem \(2013\)](#), and I cost reforms that change the shape of the schedule with a calibrated [Kleven and Waseem \(2013\)](#) notch model. To re-solve each firm’s problem under a counterfactual schedule I assign it a cost parameter consistent with its observed turnover and inputs (the residual $A_i = Y_i/X_i^\alpha$); I use this as a calibration input and do not claim to recover a policy-invariant productivity distribution or to identify structural primitives.

Three findings follow. First, the behavioural response at the threshold is small. I estimate a bunching ratio of $b \approx 0.061$ and a median local turnover elasticity of about 0.17, of a magnitude comparable to the VAT-base elasticities of 0.09 to 0.18 reported by [Liu et al. \(2021\)](#), though a distinct object. This is a static, contemporaneous statement about excess mass near the threshold and is separate from the dynamic growth-suppression channel ([Liu et al., 2024](#); [Bashir et al., 2024](#)). Because the response is small, the revenue cost of changing the threshold’s level is close to its mechanical value: my static costing of the April 2024 increase from £85,000 to £90,000 gives £175m in 2025–26 against HMRC’s published £185m, and forward moves from the £90,000 base range from a gain of £665m at £70,000 to a cost of £1,205m at £120,000, against a base of about £203.8bn.

Second, the shape and rate reforms act on the notch’s distortion in a way that a change in level does not. The notch creates a *dominated region* of width $a = T^*\tau/(1-\tau) = £21,250$ above the threshold, a range of turnover in which no firm locates. Raising the threshold relocates this region without removing it. A reduced rate shrinks it—to £9,444 at a 10% band rate and £15,000 at 15%—while a graduated taper, being continuous, removes it. Third, the reforms have comparable static revenue costs: £374m to raise the threshold to £100,000, £335m for the taper, and £357m and £178m for the 10% and 15% reduced rates. I report, for each reform, this static cost together with the portion of the dominated region it removes. The static costs are robust to the model’s smoothing parameter; the behavioural and distributional responses are not, and I report them as directional.

Related literature. The paper draws on and contributes to three strands of work. The first is the methodology of bunching and notches ([Saez, 2010](#); [Kleven and Waseem, 2013](#); [Chetty et al., 2011](#); [Kleven, 2016](#); [Best et al., 2015](#)), which provides the reduced-form estimator; I apply it and then go beyond it to cost reforms that leave no single point

of excess mass to exploit. The second is the empirical literature on the VAT registration threshold (Liu et al., 2021, 2024; Onji, 2009; Harju et al., 2019; Asatryan and Peichl, 2017; Nandi and Warwick, 2020; Bashir et al., 2024), which estimates the behavioural response and studies changes in the threshold’s level. Liu et al. (2021) is closest: it estimates bunching at the UK notch on administrative data and documents voluntary registration; I report a turnover elasticity, a distinct object from their VAT-base elasticity, and cost shape and rate reforms they do not. The international evidence (Harju et al., 2019; Onji, 2009; Asatryan and Peichl, 2017; Nandi and Warwick, 2020) notes that bunching at a VAT threshold can reflect compliance costs, firm-splitting, or reporting rather than real turnover responses. The third strand is the structural literature on size-based thresholds (Garicano et al., 2016; Gourio and Roys, 2014), which identifies policy-invariant primitives at the French 50-employee threshold and computes welfare from them; my notch model is a calibration device and makes no such claim. This paper makes two contributions. First, it provides an open, reproducible firm-level microsimulation for costing UK VAT threshold reforms, where the existing firm-level evidence on the threshold rests on restricted administrative data. Second, it costs two schedule reforms—a graduated taper and a reduced rate—that the reduced-form bunching estimator cannot price and that prior work has not costed. A welfare and marginal-value-of-public-funds accounting, revenue-neutral designs, and the treatment of voluntary registration are left to future work.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background of the UK VAT threshold. Section 3 sets out the synthetic firm microdata and its calibration to HMRC aggregates. Section 5 reports the reduced-form bunching evidence. Section 6 develops the calibrated notch model. Section 8 sets out the static costing of threshold changes and the reform menu. Section 9 concludes.

2 Institutional background

Registration and the notch. The United Kingdom levies a value-added tax at a standard rate of 20% on most goods and services. A business must register once its taxable turnover over any rolling twelve-month period exceeds the registration threshold, after which it charges VAT on its taxable sales and reclaims VAT paid on inputs. Because registration makes VAT due on the firm’s *entire* taxable turnover rather than only the amount above the cutoff, the threshold is a *notch*—a discrete jump in liability—and not a kink. The cost of crossing is largest for consumer-facing firms with limited pass-through and small reclaimable input shares, for whom the incentive is to restrain turnover and locate just below the threshold (Liu et al., 2021; Kleven and Waseem, 2013).

Value added, input reclaim, and voluntary registration. Real VAT is a tax on *value added*, not on turnover: a registered firm charges output VAT on its sales but reclaims input VAT on its purchases. A large share of below-threshold firms therefore register *voluntarily*—roughly 45% of eligible UK firms (Liu et al., 2021), driven by input-VAT reclaim and business-to-business sales—and for them the incentive to bunch is weak.

My one-input model taxes *turnover* rather than value added, so it is a *turnover-tax-notch approximation* for the consumer-facing, limited-reclaim firms for whom the bunching incentive is sharpest; it abstracts from input reclaim and voluntary registration. I treat this as a first-order scope restriction and return to it in Section 9.

Threshold history. The registration threshold was held at £85,000 from 1 April 2017 to 31 March 2024—a seven-year nominal freeze, the longest in the history of the tax—before being raised to £90,000 on 1 April 2024, with the deregistration threshold rising from £83,000 to £88,000 (HM Revenue and Customs, 2024; Seely, 2024). Because nominal turnover grows with inflation and real activity, freezing the threshold lowers it in real terms and draws an increasing number of firms towards the registration margin; this fiscal-drag mechanism was a central motivation for the April 2024 increase. My data year is 2023–24, so the prevailing statutory threshold throughout my analysis is £85,000. Figure 2 shows the pattern in the turnover distribution: the number of firms rises towards £84,000 and falls sharply above £85,000, and the projected 2025–26 distribution is more sharply bunched than the 2019–20 outturn as fiscal drag draws more firms towards the frozen threshold.

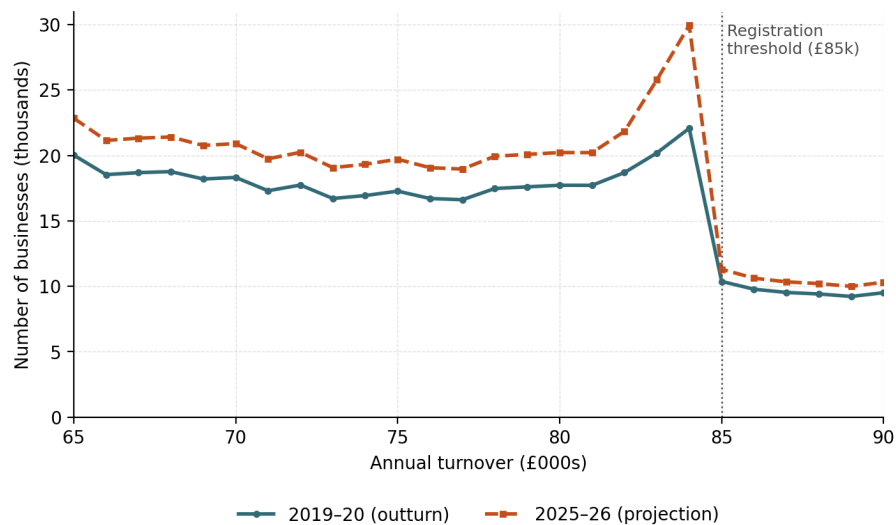


Figure 2: Bunching below the £85,000 VAT registration threshold: number of businesses by £1,000 turnover band. Source: OBR, *Economic and Fiscal Outlook March 2023*, Chart C (Open Government Licence).

The reform debate. The UK threshold is among the highest in the OECD, and there is debate over whether to raise it—to reduce the compliance burden on small firms—or lower it, to broaden the base and reduce the growth distortion at the margin. Three families of reform feature, and I evaluate all three: a change in the threshold’s *level* (the £85,000 to £90,000 increase of April 2024 and further moves around it); a change in its *shape* (a *graduated taper* that phases the effective rate in over a band, replacing the

notch with a continuous profile); and a change in its *rate* (a *reduced rate* for small firms in a band above the threshold, with the standard 20% rate retained above). A banded reduced rate introduces a smaller *secondary* notch at the band top, which the continuous taper avoids. I cost all three by the same machinery in Section 8.

3 Data

Studying firm responses at the threshold needs firm-level turnover near the cutoff, which the official statistics publish only in coarse turnover bands. I therefore build a synthetic firm population calibrated to those aggregates: it reproduces the official totals while resolving the individual firm that the authorities do not release, opening the analysis to anyone. The same calibrate-to-published-aggregates approach underlies open household-level tax-benefit microsimulation ([PolicyEngine, 2025](#)); here I apply it at the firm level. The generator takes the registration threshold as a parameter and is documented in Appendix A.1.

Construction. Index firms by i . For each sector s and ONS turnover band $b = [\underline{b}, \bar{b}]$, the *UK Business: Activity, Size and Location* table gives a firm count $N_{s,b}$, and I draw $N_{s,b}$ firms with within-band turnover

$$y_i = \underline{b} + (\bar{b} - \underline{b}) u_i + \varepsilon_i, \quad u_i \sim \mathcal{U}(0, 1), \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_b^2),$$

truncated below at zero, where ε_i smooths the within-band distribution. Each firm is given input expenditure $x_i = \rho_i y_i$, with the input-output ratio ρ_i drawn from a rescaled Beta distribution with sector-specific shifts and bounded to $\rho_i \in [0.6, 1.5]$; its value added, the net VAT base, is $v_i = y_i - x_i$. Employment is assigned from the ONS employment-band shares.

The base population reproduces the ONS structure but not the HMRC VAT-registered totals, so I re-weight it. Each firm receives a strictly positive weight $w_i = e^{\theta_i}$, and for a vector of official targets $T = (T_k)_{k=1}^K$ the weighted synthetic counterpart of target k is

$$\hat{T}_k(\theta) = \sum_i a_{ki} w_i,$$

where a_{ki} is firm i 's contribution to target k : an indicator of band or sector membership for a count target, and the firm's liability v_i for a VAT-liability target. I choose θ to minimise a symmetric relative-error loss with per-target importance weights λ_k ,

$$L(\theta) = \frac{1}{K} \sum_k \lambda_k \min \left\{ \left(\hat{T}_k / T_k - 1 \right)^2, \left(T_k / \hat{T}_k - 1 \right)^2 \right\} + \alpha \sum_i |\theta_i|,$$

the final term an L_1 penalty on the log-weights. The targets are the HMRC VAT-registered counts by turnover band and by sector, the HMRC net VAT liability by band and by sector, and the ONS employment-band totals; turnover-band targets carry the largest λ_k ,

followed by VAT liability by band. The problem is solved by gradient descent, and the symmetric loss keeps targets of very different magnitudes on a common scale.

A firm is registered with certainty when its turnover exceeds the threshold T^* . Below the threshold it registers voluntarily with probability $r = M/n_<$, the HMRC count of registered firms in the £1–threshold band, M , divided by the synthetic number of below-threshold firms, $n_<$, so the synthetic voluntary-registration rate matches the HMRC aggregate. The resulting dataset comprises approximately 2.94 million firm records, weighted to represent roughly 2.0 million UK firms; validated against the official sources along every calibrated dimension, it attains an overall calibration accuracy of about 87%, reproducing the aggregates closely while supplying the firm-level resolution they lack.

The registration threshold in the data. My data are for the 2023–24 year, in which the statutory threshold was £85,000. The firm density steps down at the threshold—from roughly 10,400 to 6,700 firms per £1,000 turnover band—reflecting the incentive to remain below the registration cutoff, with a second, smaller step at £150,000, the ceiling of the VAT Flat Rate Scheme (Figure 3). Counterfactual policies—including the £85,000 to £90,000 reform and the forward threshold sweep of Section 8—are evaluated on these firms, by re-applying the registration rule to the microdata for the static costings and by re-solving firm optimisation for the behavioural exercise (Section 8.0.1). I hold the empirical turnover distribution fixed and let behaviour respond to the reform, rather than imposing a bunching pattern at the counterfactual threshold, which would presuppose the response the analysis sets out to measure.

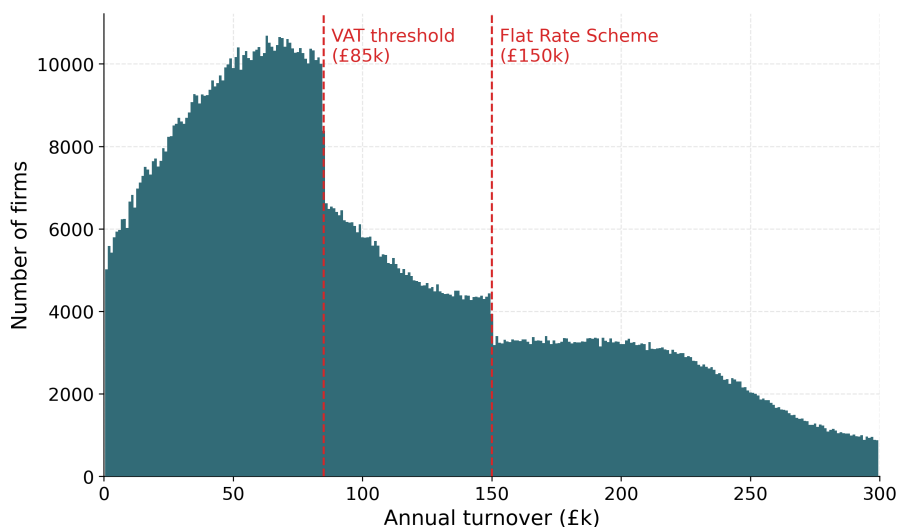


Figure 3: Synthetic firm turnover distribution (weighted firm counts, £1,000 bins), 2023–24. The density steps down at the £85,000 registration threshold and again at the £150,000 Flat Rate Scheme ceiling.

4 Static costing of threshold reforms

A static costing measures the first-order revenue effect of a reform by re-applying the tax rule to a fixed firm population, with no behavioural response. Let firm i have turnover y_i , calibrated net VAT liability v_i (output less input VAT; Section 3), and weight w_i . A firm remits VAT only once registered, that is once its turnover exceeds the threshold T , so total VAT revenue at threshold T is the weighted sum of liabilities over registered firms,

$$R(T) = \sum_i w_i v_i \mathbf{1}\{y_i \geq T\}.$$

Because turnover is held fixed, moving the threshold from T_0 to T changes revenue only through the firms whose registration status flips—those with turnover in the band between the two thresholds:

$$\Delta R = R(T) - R(T_0) = -\text{sign}(T - T_0) \sum_i w_i v_i \mathbf{1}\{T_{\min} \leq y_i < T_{\max}\},$$

where $T_{\min} = \min(T, T_0)$ and $T_{\max} = \max(T, T_0)$: lowering the threshold draws firms into the net and raises revenue, while raising it releases firms and loses revenue. The estimate is therefore the mechanical reclassification effect and excludes the behavioural response of Section 5. The calibrated weights reproduce HMRC’s net VAT-liability totals (Section 3), and I age the microdata to a given fiscal year by a cumulative nominal-growth factor. At the £90,000 threshold this puts total VAT revenue at £203.8bn in 2025–26 and £210.0bn in 2026–27.

Validation against HMRC. I first check the static pipeline against HMRC’s own costing of the April 2024 increase from £85,000 to £90,000. Because my microdata are calibrated to HMRC VAT aggregates (Section 3), a close match confirms internal consistency rather than out-of-sample accuracy. Figure 4 compares the two costings year by year: they share sign and profile throughout—a loss that narrows as fiscal drag lifts the frozen baseline towards the reformed threshold, turning to a small gain by 2028–29—and agree to within £7m–£21m per year (for 2025–26, –£175m against HMRC’s –£185m). The model reproduces both the magnitude and the turning point of the official costing.

A menu of threshold locations. Having checked the method, I use it to cost a menu of further threshold locations, taking the post-reform £90,000 threshold as the baseline. Each row of Table 1 costs a move of the threshold to a new location for 2025–26, holding the turnover distribution fixed, so the revenue change reflects only the reclassification of firms into or out of the VAT net. Because the calibration places extra weight on high-liability firms just below the threshold, I evaluate the reclassified band from a smooth extrapolation of the clean above-threshold profile rather than the raw below-threshold mass. Lowering the threshold draws firms in and raises revenue; raising it loses revenue. The schedule is close to linear—each £5,000 step changes revenue by roughly £150m–£220m, with the marginal effect growing as the threshold sweeps across denser parts of the distribution. Even a £30,000 rise to £120,000 costs about £1.2bn, under 0.6% of

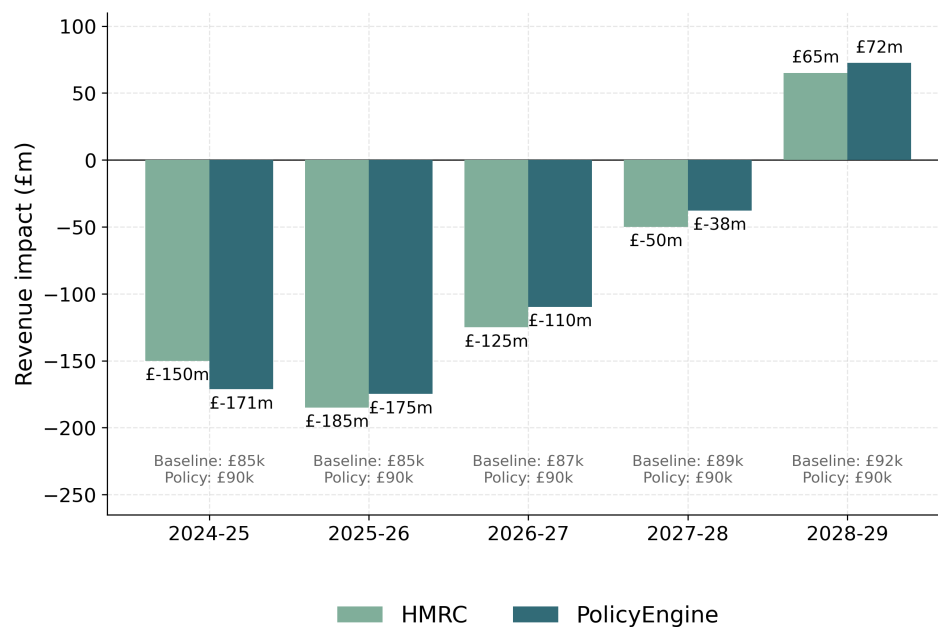


Figure 4: Anchor reform: revenue impact of raising the VAT registration threshold from £85,000 to £90,000 (the April 2024 change), my estimate versus HMRC’s published costing, 2024–25 to 2028–29. The two series share sign and profile, both turning positive in 2028–29 as fiscal drag lifts the frozen baseline above the reformed threshold. Because the microdata are calibrated to HMRC aggregates, this is a consistency check, not an out-of-sample validation.

the £203.8bn base, because the threshold governs many small firms whose individual liabilities are small. Figure 5 shows the sweep graphically. These are *static* results: they bound the first-order fiscal effect of a threshold change but abstract from the behavioural response of Section 5.

Table 1: Static revenue and VAT-paying-firm effects of moving the VAT registration threshold, 2025–26, relative to the post-reform £90,000 baseline (total VAT revenue £203.8bn). Figures are mechanical, holding firm turnover fixed (no behavioural response). This sweep is distinct from the anchor reform of Figure 4.

Threshold	Revenue change (£m)	Change (%)	VAT-paying firms (000s)
£70,000	+664.9	+0.33	+79.3
£75,000	+509.0	+0.25	+58.7
£80,000	+346.3	+0.17	+38.7
£85,000	+176.6	+0.09	+19.1
£90,000 (baseline)	0	0	0
£95,000	−183.5	−0.09	−18.6
£100,000	−374.0	−0.18	−36.8
£105,000	−571.4	−0.28	−54.5
£110,000	−775.7	−0.38	−71.7
£115,000	−987.0	−0.48	−88.5
£120,000	−1,205.1	−0.59	−104.7

5 Bunching at the VAT Threshold

This section presents reduced-form evidence of firm bunching at the VAT registration threshold, using the polynomial counterfactual-density estimator of [Saez \(2010\)](#) and [Kleven and Waseem \(2013\)](#) as applied to UK VAT data by [Liu et al. \(2021\)](#); I also draw on the synthesis of [Kleven \(2016\)](#) and the friction diagnostics of [Chetty et al. \(2011\)](#). The estimator is purely reduced-form: it fits a smooth counterfactual to the empirical turnover density and reads excess mass off the gap, without imposing any structural model of the firm. The notch mechanism that generates this mass—VAT applying to a registered firm’s entire turnover, so that locating just below the cutoff is attractive—is formalised in Section 6; here it suffices that the notch produces excess mass below the threshold and missing mass above it. I estimate bunching at the £85,000 threshold prevailing in my 2023–24 data.

5.1 Methodology

Following [Liu et al. \(2021\)](#), I group firms into fine turnover bins of £100 and estimate a counterfactual density—the distribution that would obtain absent the registration

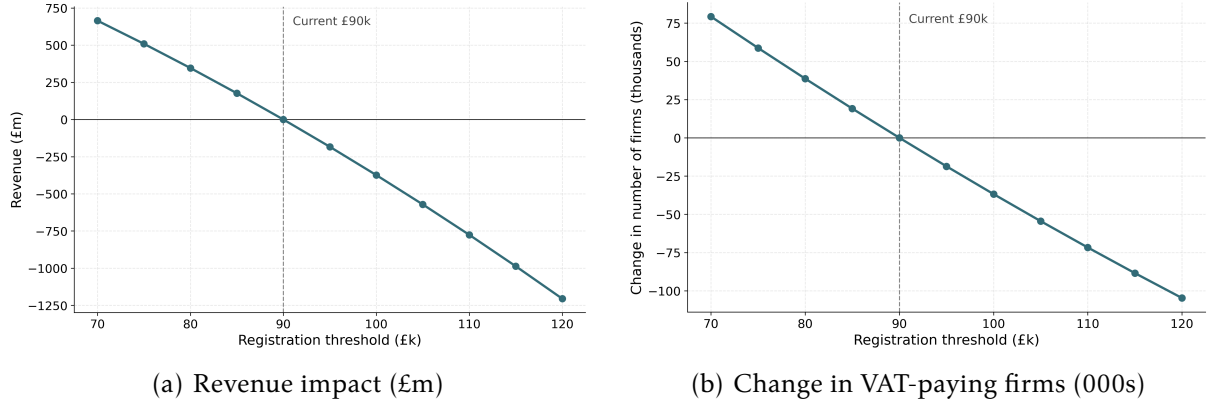


Figure 5: Static effect of moving the VAT registration threshold, 2025–26, relative to the £90,000 baseline (dashed line). Panel (a): revenue impact—lowering the threshold raises revenue, raising it loses revenue, and the schedule is approximately linear over the £70,000–£120,000 range. Panel (b): the change in the number of VAT-paying firms, tracking the local density of firms on each side of the threshold.

notch—by polynomial regression on bin counts, excluding a manipulation window around the threshold y^* :

$$c_j = \sum_{l=0}^q \beta_l (y_j)^l + \sum_{i \in [y_-^*, y_+^*]} \gamma_i 1\{j = i\} + \varepsilon_j,$$

where c_j is the number of firms in bin j , y_j is the distance of bin j from the threshold, q is the polynomial order, and $[y_-^*, y_+^*]$ is the excluded window. The fitted polynomial is then projected into the excluded region to recover the counterfactual density $g(y)$. Excess bunching is the integrated difference between the observed and counterfactual densities just below the threshold,

$$B(y^*) = \int_{y^* - \Delta y^*}^{y^*} g(y) dy \approx g(y^*) \Delta y^*,$$

the approximation holding when g is locally smooth. The bunching ratio $b(y^*) = B(y^*)/g(y^*) \approx \Delta y^*/y^*$ is the width of the bunching region expressed as a fraction of the threshold—a normalised *ratio*, not an elasticity. Translating the excess mass into a behavioural elasticity requires the structural step of Section 6; the estimator here delivers the raw geometry of the response.

5.2 Results

For the 2023–24 data, measured at the £85,000 threshold, my *preferred* specification imposes the mass-conservation restriction that excess mass below the threshold equal missing mass above it, locates the marginal buncher endogenously, and bootstraps the

sampling distribution (Appendix A.3). It yields a bunching ratio $b = 0.061$ and excess mass of 9,160 firms; full inference (bootstrap standard errors and the mass-conservation constraint) is reported in Section 7 and Appendix A.3. Mass conservation matters: an unconstrained polynomial fit (reported in Figure 6) implies excess against missing mass that grossly violate adding-up, whereas the constrained estimator reconciles the two. In all specifications the observed density displays clear excess mass in the £100 bins immediately below the threshold and a corresponding deficit just above it, the signature of a notch-induced behavioural response. A non-trivial share of firms near the threshold thus adjust their turnover to avoid registration; the behavioural turnover elasticity that this implies (local median ≈ 0.17) is recovered in Section 7 and sits within the range Liu et al. (2021) report for the UK VAT base.

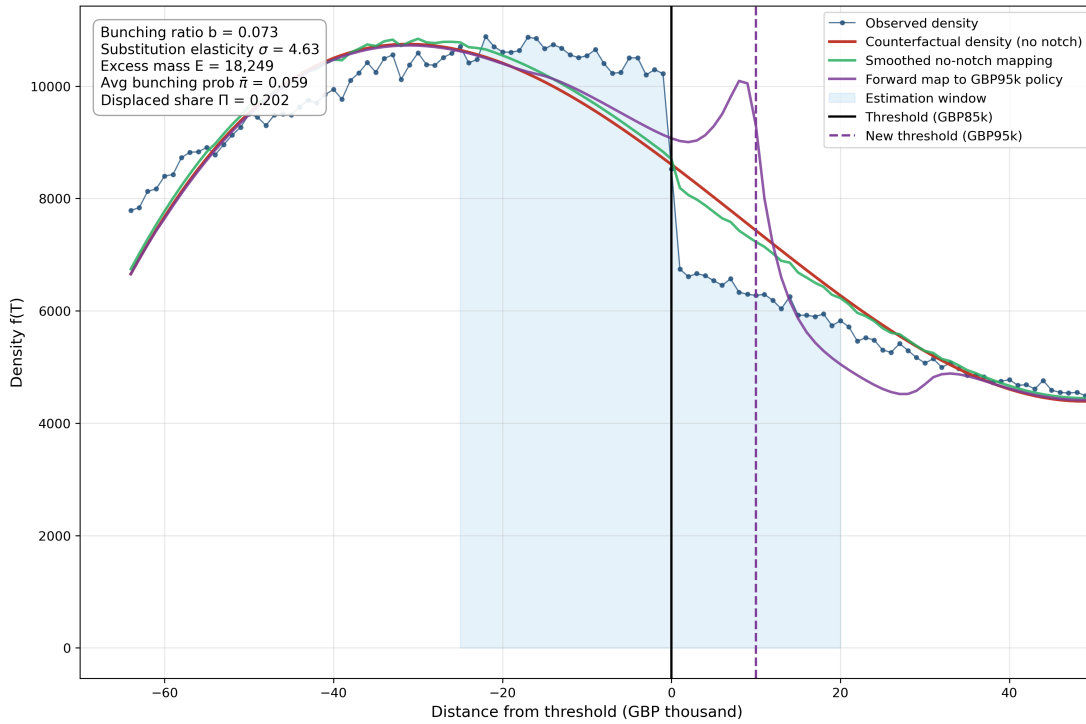


Figure 6: Bunching analysis at the £85,000 threshold (2023–24 data year). The figure shows the observed firm density, the estimated polynomial counterfactual density fitted outside the manipulation window, and the implied distribution under a forward-mapped £95k new-policy threshold. The estimates shown here, from the plotting pipeline, are a bunching ratio $b = 0.073$ (a width ratio, not an elasticity), substitution elasticity $\sigma = 4.63$, and excess mass = 18,249 firms. My *preferred* estimates, from the mass-conserving estimator with bootstrap inference (Appendix A.3), are $b = 0.061$ and $\sigma = 5.10$; its unconstrained variant gives $b = 0.054$.

5.3 From reduced-form evidence to structure

These estimates quantify the aggregate response at the prevailing threshold but do not identify the underlying firm-level productivity distribution. For a move of the threshold itself the reduced-form counterfactual already suffices; what it cannot price is a change to the *shape* of the schedule, which has no single point of excess mass to forward-map. The calibrated model of the next section assigns each firm a cost parameter consistent with its observed turnover and re-solves its choice under any schedule, which is what lets me cost the taper and reduced-rate reforms that follow.

6 A Calibrated Notch-Bunching Model of Firm Turnover

The reduced-form evidence in Section 5 establishes that firms respond to the VAT registration notch, but it cannot price reforms the data never observed—a relocated threshold, a reduced small-firm rate, or a tapered schedule. To do that I calibrate a notch-bunching model of firm turnover, following the framework of Kleven and Waseem (2013), Saez (2010), and Kleven (2016), as applied to UK VAT by Liu et al. (2021). The UK registration threshold is a *notch*, not a kink: once turnover crosses the threshold the firm becomes liable for VAT on its *entire* turnover, not merely on the increment above it. I model this notch exactly. The behavioural elasticity that governs the response, and its calibration, are reported in Section 7.

This is a calibrated policy model, not a structural identification of firm primitives. I assign each firm a cost/productivity parameter consistent with its observed turnover and inputs and re-solve its problem under counterfactual schedules; I do *not* claim to recover a policy-invariant productivity distribution or to identify structural primitives. Papers that do identify such primitives at a size threshold—Garicano et al. (2016) and Gourio and Roys (2014)—instrument or structurally restrict the technology; I differ precisely by not making that claim.

Scope: a turnover-tax notch. The model taxes *turnover* through a one-input technology. Real VAT is levied on *value added* with input-tax reclaim, and roughly 45% of below-threshold firms register voluntarily to recover input VAT (Liu et al., 2021). My model abstracts from input reclaim and voluntary registration: it is a turnover-tax-notch approximation. This is a scope limitation I state up front; it bounds the welfare claims I make and is revisited in Section 9.

6.1 The firm’s problem with a VAT notch

Each firm i chooses turnover y_i to maximise after-tax profit. Let $c_i(y_i)$ denote the cost of producing turnover y_i , T^* the registration threshold, and $\tau_{\max}$ the standard VAT rate. Because VAT applies to the firm’s entire turnover once it registers, after-tax revenue is

discontinuous and profit is

$$\pi_i(y_i) = \begin{cases} y_i - c_i(y_i), & y_i < T^* \quad (\text{unregistered}), \\ (1 - \tau_{\max}) y_i - c_i(y_i), & y_i \geq T^* \quad (\text{registered}). \end{cases} \quad (1)$$

At T^* after-tax revenue falls discretely by $\tau_{\max} T^*$: a firm that nudges its turnover from just below to just above the threshold loses $\tau_{\max} T^*$ of net revenue on turnover it was already earning (about £17,000 at my parameters). This discrete drop—absent from a kink, where only the *marginal* rate changes—is the defining feature of a notch and the source of the bunching documented in Section 5.

6.2 A Cobb–Douglas cost specification

I specify cost through a one-input Cobb–Douglas technology relating turnover Y_i to input expenditure X_i ,

$$Y_i = A_i X_i^\alpha, \quad c_i(y_i) = \left(\frac{y_i}{A_i} \right)^{1/\alpha}, \quad (2)$$

where $A_i > 0$ is a firm-specific cost/productivity parameter and $\alpha > 0$ a common curvature parameter. The implied marginal cost is

$$\frac{\partial c_i(y_i)}{\partial y_i} = \frac{1}{\alpha} \frac{c_i(y_i)}{y_i} = \frac{1}{\alpha} \frac{X_i}{Y_i}, \quad (3)$$

identified from the firm’s input–output ratio in the data. The curvature parameter is calibrated to $\alpha \approx 0.989$ (Section 7), i.e. near-constant returns, so marginal cost is close to X_i/Y_i .

6.3 The dominated region

The notch creates a range of turnover just above T^* that *no* firm chooses, because every point in it yields strictly lower profit than locating at T^* . A firm at the threshold earns net revenue T^* ; a registered firm at $T^* + a$ earns $(1 - \tau_{\max})(T^* + a)$. Equating the two (the cost difference across this narrow band is negligible), the upper edge of the dominated region is

$$(1 - \tau_{\max})(T^* + a) = T^* \implies a = T^* \frac{\tau_{\max}}{1 - \tau_{\max}}. \quad (4)$$

This is the Kleven–Waseem dominated-region width (Kleven and Waseem, 2013). At my parameters it gives $a = £21,250$, so the dominated region is $(T^*, T^* + a) = (£85,000, £106,250)$. This $\sim £21\text{k}$ band—an order of magnitude wider than the $\sim £1\text{k}$ region implied by a smoothed *kink* approximation—is the load-bearing analytic object of the paper: it is exact, independent of α and of any smoothing parameter, and it underwrites the efficiency ranking of reforms in Section 8.0.1. I return to the kink contrast in Appendix A.2.

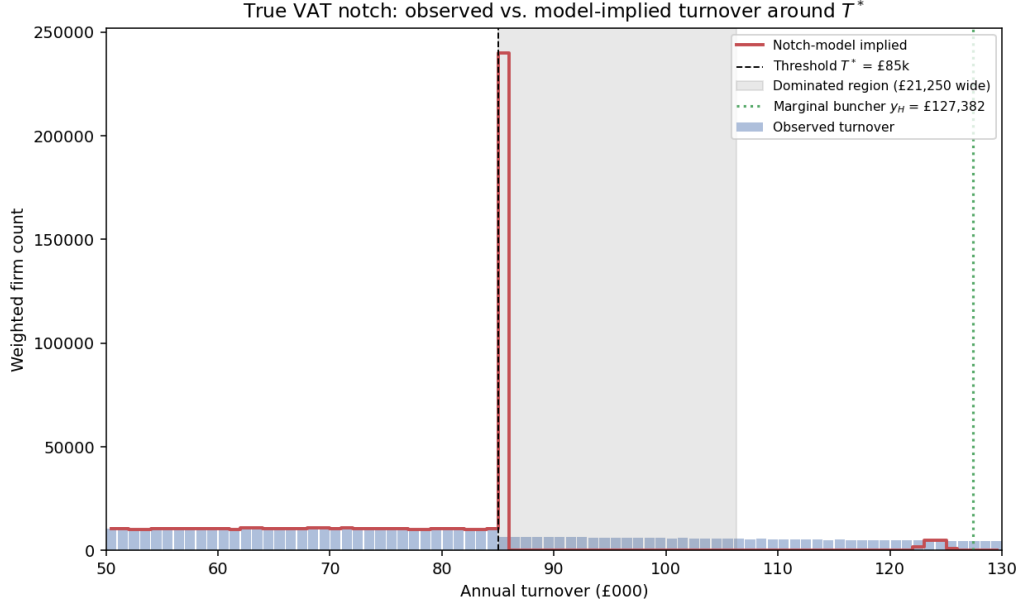


Figure 7: The VAT notch at $T^* = £85,000$. The model predicts a spike of bunchers at the threshold, an empty dominated region of width $a = £21,250$ (shaded), and registered firms locating above the marginal buncher $y_H = £127,382$ (at the median turnover elasticity $e = 0.17$). The observed turnover distribution is shown in blue.

6.4 The marginal buncher

Firms whose frictionless optimum lies above T^* choose between *bunching* at the threshold (remaining unregistered) and *registering* beyond the dominated region. The marginal buncher y_H is exactly indifferent between the two; all firms with frictionless optimum in $(T^*, y_H]$ bunch at T^* , while firms above y_H register and jump past the dominated region. The width of the bunching segment $\Delta y^* = y_H - T^*$ is the structural counterpart of the excess mass measured in Section 5, and it pins down the behavioural turnover elasticity e , exactly as in Kleven and Waseem (2013) and Saez (2010).

Because $\alpha \approx 0.989$ implies near-constant returns, I evaluate the marginal buncher in the standard Kleven–Waseem iso-elastic representation, in which the behavioural elasticity e governs the response, rather than through the Cobb–Douglas branch optimum. This gives $y_H = £127,382$ ($\Delta y^*/T^* = 0.50$) at the median elasticity $e = 0.17$, rising to $y_H = £143,527$ at the mean elasticity $e = 0.32$; in both cases y_H lies well above the dominated region, as it must. Figure 7 displays the resulting structure—a spike of bunchers at T^* , the empty dominated region, and registered firms beyond y_H —against the observed turnover distribution.

6.5 The calibration input and forward solution

To re-solve the firm’s problem under a counterfactual schedule I need a firm-level cost parameter consistent with each firm’s observed allocation. I take it as the production residual

$$A_i = \frac{Y_i}{X_i^\alpha}, \quad (5)$$

using only observed turnover Y_i , inputs X_i , and the calibrated α . This residual is a *calibration input*, not an identified structural primitive: it is whatever cost level rationalises the firm’s observed turnover and inputs at the £85,000 schedule, and I hold it fixed when I change the schedule. As noted above, identifying such a parameter structurally is the contribution of [Garicano et al. \(2016\)](#) and [Gourio and Roys \(2014\)](#); I do not claim it.

Forward solution. Holding $\{A_i\}$ fixed, I re-solve each firm’s problem with the schedule replaced by a counterfactual—a relocated threshold, a changed rate, or a tapered schedule. For this computation I represent the schedule with a smooth (sigmoid) effective rate, $\tau(y) = \tau_{\max}/(1 + e^{-k(y-T^*)})$, calibrated so that its steepness k reproduces the dominated region of Section 6.3 (Appendix A.2). Because the near-constant-returns technology makes a differentiable first-order condition ill-conditioned, I locate each firm’s optimum by maximising after-tax profit directly over a bounded $\pm£20,000$ grid around its observed turnover rather than by root-finding. The bounded window is a modelling assumption, justified by the localisation of the realised response (Section 8.1); I flag that it mechanically limits how far firms reposition. The smooth schedule and the grid search are the *computational implementation* of the notch model, not a competing specification. These counterfactuals are developed in Section 8; they are distinct from the reduced-form density counterfactual introduced next.

6.6 A reduced-form counterfactual density

To quantify the distortion induced by the notch I also require the turnover distribution that would obtain in its absence. Rather than re-solving the firm’s problem, this object estimates the smooth, undistorted size distribution directly from the data, using the same polynomial counterfactual as the reduced-form bunching estimator of Section 5. Measuring each observation by its distance from the cutoff $y_{\text{dist}} = T - T^*$, I fit a third-order polynomial by ordinary least squares to the data outside a window $[T^* - W, T^* + W]$ with $W = 15k$,

$$f^{\text{cf}}(T) = \beta_0 + \beta_1 y_{\text{dist}} + \beta_2 y_{\text{dist}}^2 + \beta_3 y_{\text{dist}}^3, \quad (6)$$

and project it back into the excluded region. The gap between the observed density and (6)—excess mass below T^* and missing mass above it—measures the behavioural response, in the reduced-form tradition of [Liu et al. \(2021\)](#); it is not the structural re-optimisation of Section 8.0.1.

7 Calibration and Estimation

This section describes how the calibrated notch model of Section 6 is taken to the data. I proceed in three steps: estimate the curvature parameter α of the cost specification from firms away from the notch; assign each firm a cost/productivity parameter A_i as a calibration input; and extract the headline behavioural elasticity, reporting bootstrap standard errors for the bunching estimates.

7.1 Estimating the cost-function curvature

The cost specification is Cobb–Douglas, $Y = A X^\alpha$, so that in logs $\ln Y = \ln A + \alpha \ln X$. I estimate α by a weighted regression of $\ln Y$ on $\ln X$ across firms. To avoid contaminating the estimate with the behavioural response the model is meant to explain, the estimation sample excludes firms in the manipulation region around the data-year threshold $T^* = £85,000$, retaining firms with turnover below £75,000 or above £95,000, which face no marginal registration incentive.

The estimated curvature is $\alpha \approx 0.989$ on the £85,000 synthetic data (precisely 0.9888; an earlier hardcoded value of 0.9864 is no longer used)—near-constant returns.¹

7.2 Assigning the firm-level cost parameter

Given α , I assign each firm the cost/productivity parameter $A_i = Y_i/X_i^\alpha$, the production residual implied by its observed turnover Y_i and inputs X_i . This is a *calibration input* for the forward re-solution of Section 6, not a claim of structural identification: it is whatever cost level rationalises the firm’s observed allocation at the £85,000 schedule, held fixed when the schedule changes. I do not pre-process turnover with any “un-bunching” rule or inject uncertainty before forming A_i , so the same residual is carried unchanged into the forward solution. The assignment is available for the 91.5% of firms with strictly positive observed inputs and turnover; the values are heterogeneous and broadly increasing in firm size.

7.3 The headline behavioural elasticity

The behavioural object of interest is the elasticity of reported turnover with respect to the net-of-tax rate. I obtain it by perturbing the VAT rate by a small $\Delta\tau$, re-solving each

¹This is an un-instrumented OLS slope of $\ln Y$ on $\ln X$ with a single input, with no control-function or proxy-variable correction (Levinsohn–Petrin, Olley–Pakes, ACF). If more productive firms purchase more inputs, the estimate is biased upward towards unity, so any object that turns on the gap between α and 1 is fragile. I therefore route the load-bearing objects away from α : the dominated region is the exact Kleven–Waseem width, the marginal buncher is evaluated in the iso-elastic representation with the behavioural elasticity (Section 6.4), and the schedule-reform revenue counterfactuals turn on the schedule and turnover, not on the precise value of α .

firm’s problem, and computing

$$\varepsilon = \frac{\ln y^*(\tau_{\max} + \Delta\tau) - \ln y^*(\tau_{\max})}{\ln(1 - \tau_{\max} - \Delta\tau) - \ln(1 - \tau_{\max})}.$$

Computed firm by firm near the threshold, this local turnover elasticity has a median of approximately **0.17** and a weighted mean of approximately 0.32 (the mean pulled up by a thin tail of very responsive firms). As a loose magnitude check, the median is of the same order as the 0.09–0.18 VAT-*base* elasticity reported by [Liu et al. \(2021\)](#) for the UK; this is only a magnitude comparison, since their object is a value-added (base) elasticity for which they decline to back out a turnover elasticity, whereas mine is a turnover elasticity. *This is the elasticity I carry into the policy analysis.* Alongside it I report the descriptive bunching ratio $b = 0.061$ (SE 0.005)—the excess-mass *width* $\Delta y^*/y^*$, not an elasticity.²

The mass-conservation constraint (Appendix A.3) raises b (from 0.054) and the excess mass E (from 8,179 to 9,160) relative to the unconstrained fit, and forces excess and missing mass into agreement ($E = 9,160$ versus $\Delta_R = 9,938$, against 8,179 versus 17,792 unconstrained); the implied displaced share $\Pi = 0.220$ is robust even where the wedge level is not. The sigmoid steepness k does not affect these data-side magnitudes but governs the width of the structural schedule’s dominated region; I set $k = 0.001$, whose implied band (£19.7k) is closest to the exact [Kleven and Waseem \(2013\)](#) notch region $T^*\tau/(1 - \tau) = £21.25\text{k}$. Re-solving the calibrated model reproduces the observed near-threshold distribution closely, but this is true essentially by construction (the model is fitted to that distribution) and I do not treat the fit as an independent result.

8 Results: Costing the Reform Menu

This section reports the policy payoff of the microsimulation. The headline deliverable is a *menu* of registration-threshold reforms—raising the threshold (a *location* move), a graduated *taper*, and a banded *reduced rate*—costed on a common static basis and then *ranked by efficiency*: how much of the notch’s distortion each reform removes relative to its fiscal cost. The efficiency ranking (Table 3) is the paper’s central policy result, and it rests on a robust, analytic object—the notch’s dominated region—rather than on any fragile simulation. I build to it in stages: the modest, localised behavioural response from the calibrated model; the graduated taper; the reduced rate; and the efficiency ranking that orders the menu. The static costing of location moves, which serves as the cost benchmark, is in Section 4.

²The bunching estimator also implies a substitution (CES-wedge) elasticity $\sigma \approx 5.10$ (SE 0.12), which reads the full excess span as a response; it is not comparable to the local turnover elasticity and is not used in the policy analysis. See Appendix A.3.

8.0.1 Counterfactual policy reforms

This paper delivers two kinds of counterfactual. The first kind is a change to the *location* of the single VAT registration threshold. I organise these around two objects. The first is the *anchor reform*: the actual increase of the registration threshold from £85,000 to £90,000 that took effect on 1 April 2024. This is a real, recent policy change, and I benchmark my static costing of it directly against HM Revenue and Customs’ own published costing (Section 4). The second is a *forward-looking sweep* of alternative threshold locations spanning £70,000 to £120,000, costed from the post-reform £90,000 baseline. The anchor reform tells me whether the model reproduces a costing I can check; the sweep then maps out the fiscal menu of further location moves.

The second kind is a change to the *shape* of the schedule: Section 8.0.4 delivers a *graduated taper* that replaces the £85,000 notch with an effective rate that phases linearly from 0% at £85,000 to the full 20% at £105,000. A schedule-shape change has no single point of excess mass that a reduced-form bunching estimator can forward-map; pricing it requires re-solving each firm’s choice under a new, continuous tax function, which the calibrated model lets me do.

Alongside these threshold-location reforms, the microsimulation assigns each firm a cost parameter $\{A_i\}$ consistent with its observed turnover and inputs (Section 6); I make no claim to identify it as a structural primitive. Holding $\{A_i\}$ fixed, I replace the tax schedule $\tau(\cdot)$ with a counterfactual policy, re-solve each firm’s choice for a new optimal turnover y_i^{new} , and re-aggregate the weighted choices into a counterfactual turnover distribution.

8.0.2 What the calibrated model adds

For the revenue cost of a pure threshold-*location* change, a static mechanical calculation already captures the first-order fiscal effect—the close HMRC consistency check in Section 4 comes from that calculation—so I do not lean on the calibrated model there. Its distinctive payoff is to price the reform a reduced-form estimator cannot: a change to the *shape* of the schedule, which has no single point of excess mass to forward-map. I exploit this in Section 8.0.4 to replace the notch with a graduated taper. Along the way the model also simulates the full firm-size *distribution* under unobserved thresholds and separates optimisation from frictions, neither of which the reduced form isolates. I exploit it again for a second schedule-shape reform—a banded reduced rate for small firms (Section 8.3). A *sector-specific* split rate for labour-intensive services is feasible with the same machinery but is left to future work (Section 9).

8.0.3 Frictions matter: a dynamic re-optimisation check

Before pricing the taper, I verify that the calibrated behavioural rule, not frictionless optimisation, is the right tool—evidence that disciplines the model. I re-solve the firm-size distribution under a threshold move (£85,000 to £95,000) under two assumptions. In the *Pure-FOC* variant every firm re-optimises fully to its new FOC solution, collapsing the

old mass and re-forming it as a sharp double spike at the new cutoff. The *FOC+Uncertainty* variant adds a threshold-aware perturbation, $y_i^{\text{unc}} = y_i^{\text{FOC}} (1 + \varepsilon_i)$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, $\sigma_\varepsilon = 0.04$, capturing the frictions and imperfect foresight that keep firms off the kink; this replaces the double spike with a rounded hump and reproduces the observed pattern that only a minority reposition. The frictionless variant materially overstates the response—the quantitative gap (mean turnover £1,488k versus actual £1,130k) is stated in Section 8.1—so FOC+Uncertainty is the variant I carry into the taper. Figure 8 contrasts the two against the actual distribution.

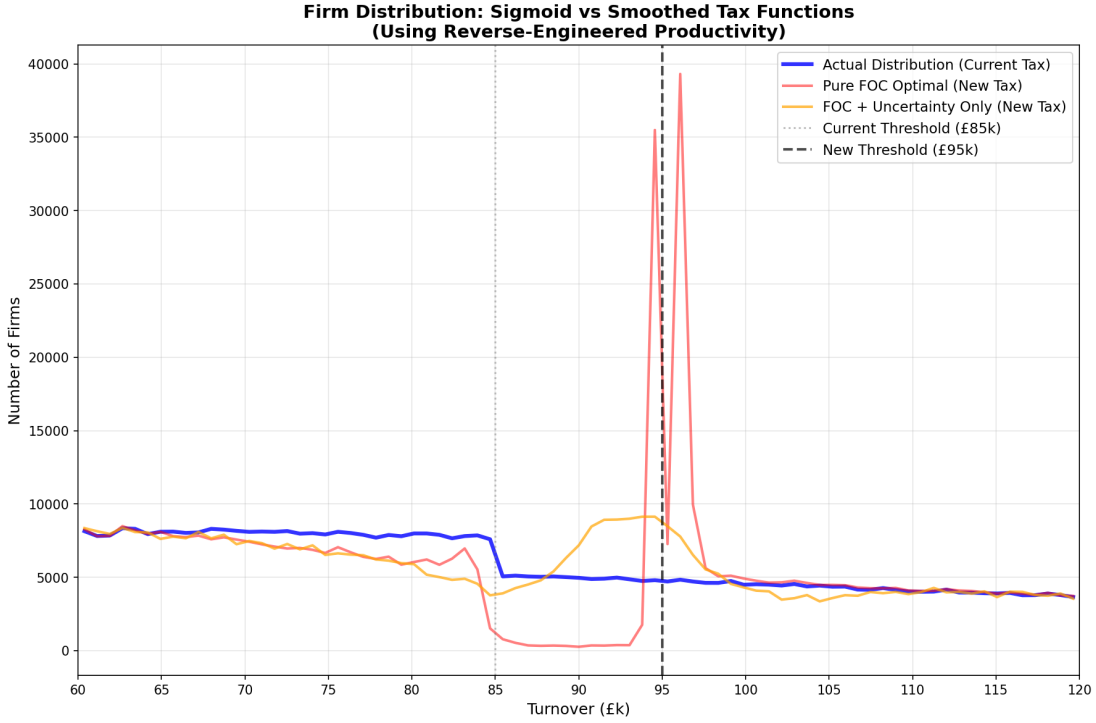


Figure 8: Dynamic re-optimisation under a new £95,000 threshold, applied to the £85,000-era data. The *Actual* distribution shows the density step at the old £85,000 threshold. The *Pure-FOC* variant, in which every firm re-optimises fully and frictionlessly, collapses the old mass and re-forms it as a sharp double spike at the new £95,000 threshold, overstating the realised response. The *FOC+Uncertainty* variant adds a threshold-aware perturbation $\varepsilon \sim \mathcal{N}(0, 0.04^2)$, replacing the double spike with a smoothed hump around the new threshold that matches the realised, partial nature of firm responses.

8.0.4 A graduated taper: the delivered schedule-shape reform

The dynamic experiment above still concerns a threshold-*location* move, for which the reduced form is a sufficient tool. I now use the calibrated model for the task it alone can perform: pricing a change to the *shape* of the schedule. I replace the notch with a

graduated taper. Let T_0 denote the lower edge of the taper band (the current registration threshold), T_1 its upper edge, and $\tau_{\max}$ the standard VAT rate. Under the taper a firm with turnover y faces an effective rate

$$\tau_{\text{eff}}(y) = \begin{cases} 0, & y < T_0, \\ \tau_{\max} \frac{y - T_0}{T_1 - T_0}, & T_0 \leq y \leq T_1, \\ \tau_{\max}, & y > T_1, \end{cases} \quad (7)$$

so the effective liability phases linearly from 0 at T_0 to the full rate $\tau_{\max}$ at T_1 . I implement the taper with $T_0 = \text{£}85,000$, $T_1 = \text{£}105,000$ (a $\text{£}20,000$ band) and $\tau_{\max} = 0.20$. The taper eliminates the discrete jump in tax liability that creates the notch: a firm that crosses $\text{£}85,000$ no longer faces VAT on its entire value added at once, only on the marginal slice implied by its position in the band.

To price this reform I hold the calibrated firm cost parameters $\{A_i\}$ fixed and re-optimize each firm's profit under the tapered $\tau(\cdot)$. Because near-constant returns make the unconstrained optimum a knife edge, I re-optimize over a bounded neighbourhood of each firm's observed turnover, consistent with the partial, localised responses documented in Section 8.1. The crucial difference between the two schedules is the *local tax wedge* $y\tau'(y)$: under the notch this wedge is large and concentrated at $\text{£}85,000$ (it is what drives firms to bunch just below); under the taper it is small and constant across the band, so the incentive to suppress turnover largely disappears. The reduced-form bunching estimator cannot evaluate this reform, because once the notch is replaced by a smooth schedule there is no single point of excess mass to forward-map.

The quantitative results, distributional effect, and figure for the taper are reported in Section 8.2.

8.0.5 Static versus behavioural evaluation

I distinguish the dynamic, behavioural simulation above from a *static* counterpart. In the static approach, firm turnover is held fixed and the new schedule applied mechanically, so revenue changes reflect only reclassification across the threshold; this benchmark underlies both the anchor-reform check and the threshold sweep. The behavioural FOC+Uncertainty simulation instead captures the second-order offset from re-optimisation and is the relevant object for how a reform reshapes the firm-size distribution. The revenue, firm-count, and distributional results for the threshold-location reforms are reported in Section 8.

8.1 The behavioural response is modest and localised

The static figures of Section 4 assume firms do not respond to the changed schedule. The calibrated model relaxes that assumption and quantifies how much firms actually reposition; the central finding is that the realised response is small. The headline behavioural object is the local turnover elasticity of Section 7 (median ≈ 0.17), broadly

in the range of the UK VAT-base elasticity of [Liu et al. \(2021\)](#). The calibrated model is fitted to reproduce the observed turnover distribution, so its close agreement with the data (to within about 0.1% on the bunching mass) is true by construction and I read it only as an internal-consistency check, not a result. The informative contrast is with the frictionless benchmark, and unlike the fit it is *not* mechanical: a Pure-FOC variant in which every firm re-optimises fully overstates the response (Figure 8), implying a mean turnover of £1,488k against an actual £1,130k.

Two regularities sharpen the interpretation. The response is *partial*: only about 15% of near-threshold firms adjust, the rest held in place by optimisation frictions and the discrete character of registration ([Chetty et al., 2011](#)). And it is *localised*: firms that respond reposition within roughly £20,000 of the threshold. Realised behaviour at the notch is therefore muted, so the aggregate behavioural offset to a threshold change is small relative to the mechanical reclassification effect, and the static figures of Table 1 are a reasonable first approximation to the cost of a threshold move.

8.2 The graduated-taper reform

I now report the schedule-shape reform of Section 8.0.4: replacing the £85,000 notch with a graduated taper whose effective rate phases linearly from 0% at £85,000 to the full 20% at £105,000. This is a counterfactual the reduced-form estimator cannot price, since a smooth schedule leaves no single point of excess mass to forward-map; I re-solve each firm’s problem under the tapered schedule, holding the calibrated cost parameters fixed, and compare with the model’s £85,000-notch baseline.

Efficiency case (analytic, robust). The taper’s case rests on an exact, parameter-free property of the notch geometry. The £85,000 notch creates a *dominated region* $(T^*, T^* + a) = (£85,000, £106,250)$ of width $a = T^* \tau_{\max} / (1 - \tau_{\max}) = £21,250$ (Section 6.3) that no firm optimally occupies—a pure misallocation zone—together with a discrete drop of $\tau_{\max} T^* \approx £17,000$ in net revenue at the threshold. A continuous taper has *no* dominated region and *no* discrete tax wedge: marginal net revenue declines smoothly through the band rather than off a cliff, so the discrete incentive to suppress turnover disappears. This follows from (4) and the firm’s problem (1), with no dependence on the smoothing parameter k , the returns-to-scale parameter α , or any simulated re-optimisation.

Revenue. Against a baseline VAT base of £179.3bn (the £85,000-notch year), the taper reduces revenue by £335m on a *static* basis (firms in the band remit a tapered rather than a full rate, turnover held fixed). I treat this static figure as the robust object. The behavioural estimate—a larger loss of about £889m once firms re-optimize and expand into the low-rate band—is reported only as a directional indication: it is sensitive to the smoothing parameter k and the re-optimisation window and I do not headline it. The reform affects about 227,000 firms, of which roughly 126,000 (125,608 weighted) face a lower effective rate by sitting in the £85,000–£105,000 band.

Sensitivity. The revenue figures turn on the taper schedule and turnover, not on the smoothed notch baseline, and are stable across k and the window. The *simulated* change

in near-threshold mass is not: the model’s notch-baseline bunching depends on k (a steep k concentrates a spike the taper then cuts; at my calibrated $k = 0.001$ the simulated notch barely bunches, so the sign of the implied change reverses). I therefore rest the distributional argument on the exact, k -free analytic mechanism above rather than on a simulated magnitude.

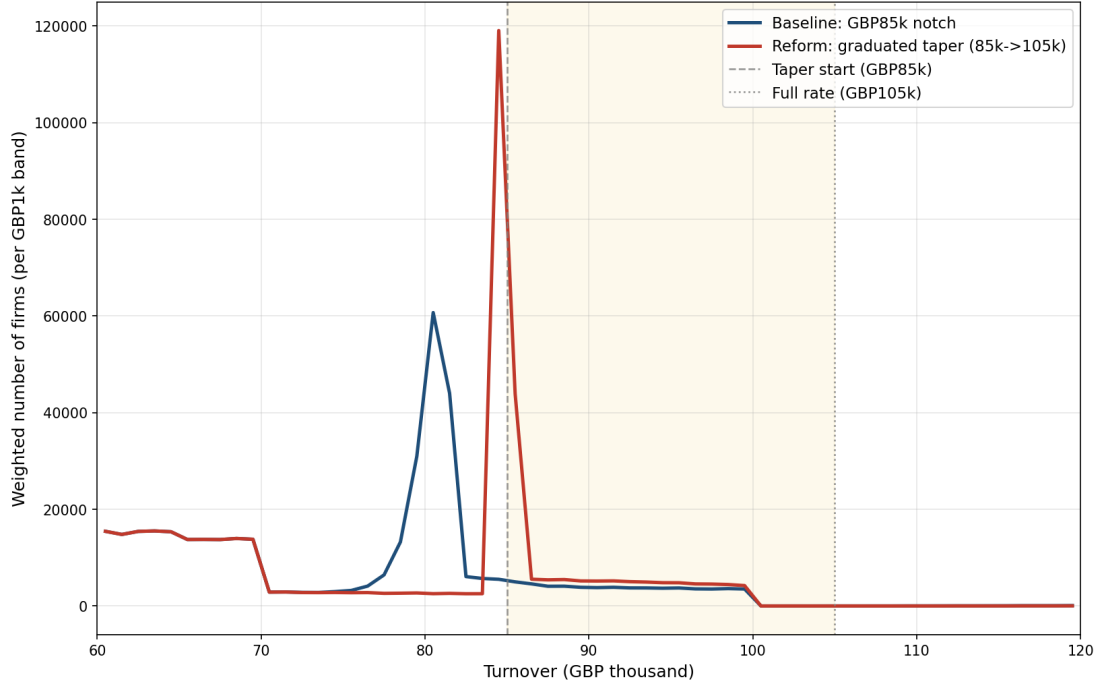


Figure 9: Graduated-taper reform: firm-size distribution under the £85,000 notch (baseline) versus the graduated taper that phases the effective VAT rate linearly from 0% at £85,000 to 20% at £105,000 (shaded band). The taper replaces the discrete notch—and its dominated region (£85,000, £106,250)—with a smooth schedule, removing the discrete incentive to suppress turnover. The precise simulated near-threshold mass is sensitive to k and the re-optimisation window, so I report the robust revenue effect and the analytic efficiency mechanism rather than a distributional magnitude.

8.3 A reduced rate for small firms

The second schedule reform changes the *rate* on a band: firms in the £85,000–£105,000 band pay a reduced rate $\tau_{\text{low}} \in \{10\%, 15\%\}$ in place of the standard 20%, with 20% retained above £105,000. As with the taper, I re-solve each firm’s problem under the new banded schedule and compare with the £85,000-notch baseline.

Revenue. Against the same £179.3bn baseline, the banded reduced rate reduces revenue, *statically*, by £357m at $\tau_{\text{low}} = 10\%$ and by £178m at $\tau_{\text{low}} = 15\%$, with about 125,600 firms in the band facing the lower rate. I report these static figures as the robust object.

The behavioural estimates are fragile to k and the window—the apparent behavioural *gain* at $\tau_{\text{low}} = 10\%$ (around +£723m) is an artefact of the bounded re-optimisation and the secondary notch below, not a robust prediction—so I mention them only as directional.

Distortion case (analytic, robust). The notch’s discrete penalty τT^* and dominated region $a = T^*\tau/(1 - \tau)$ (Section 6.3) both shrink with the rate. At $T^* = £85,000$, lowering the rate from 20% (penalty £17,000, region £21,250) to 15% cuts the region to £15,000, and to 10% to £9,444. A lower band rate mechanically shrinks the misallocation zone, and with it the incentive to bunch.

Secondary notch. The banded rate buys this at a cost the taper avoids: because the rate steps back to 20% at £105,000, the reform introduces a *smaller secondary notch* at the band top (a step of $20\% - \tau_{\text{low}}$ rather than the full 20%). The notch is attenuated and relocated, not eliminated. The taper’s continuous phase-in leaves no such step.

Scope: banded, not blanket. I price only the *banded* small-firm reduced rate. A *blanket* reduced rate on the whole registered base is an economy-wide policy—of the order of –£45bn at a 15% standard rate and –£90bn at 10%—whose incidence runs through consumer pass-through and lies outside this partial firm model.

8.4 Ranking the menu by efficiency

The reforms cost broadly the same in static revenue (Table 2), so the policy question is which buys the most distortion reduction per pound. Table 3 ranks the menu on exactly that basis, and it is the paper’s central policy result.

The distortion measure is robust and analytic. The notch’s dominated region ($T^*, T^* + a$), $a = T^*\tau/(1 - \tau)$, is a pure misallocation zone: no firm optimally locates there, so its width is a clean, k -free, α -free index of the discrete distortion the notch imposes. For each reform I ask how much of this zone it removes. Raising the threshold to £100,000 simply *relocates* the same £21,250 zone to the new cutoff, removing essentially none of it at a static cost of £374m. A reduced rate *shrinks* the zone: to £15,000 at 15% (removing £6,250) and to £9,444 at 10% (removing £11,806). The graduated taper *eliminates* it: a continuous schedule has no dominated region, so the whole £21,250 zone is removed—and at a static cost (£335m) below that of the threshold move.

The ranking carries a clear message: *schedule reforms dominate threshold-location moves on efficiency*. A location move is fiscally comparable to the schedule reforms but buys almost no distortion reduction—it merely transports the misallocation zone to a new threshold. The reduced rate shrinks the zone, more so the lower the rate. The graduated taper is the efficiency winner: it removes the entire misallocation zone, and does so at a lower static cost than raising the threshold. The reduced rate sits between the two on the same frontier, simpler than the taper but reintroducing a smaller secondary notch at the band top that the taper’s continuous phase-in avoids.

The cross-check in Table 2 is reassuring: the flat-10% band (–£357m) costs almost exactly what the taper costs (–£335m), because a $0\% \rightarrow 20\%$ linear phase-in averages

Table 2: Reform-menu comparison: static, behaviour-free revenue effect of the four reforms at comparable scale, relative to the £85,000-notch baseline (VAT base £179.3bn for the schedule reforms; the threshold move is measured against the post-reform £90,000 baseline of Table 1). Each reform pulls a different lever—location, schedule shape, or rate. Figures are static; behavioural estimates are reported and caveated as directional in the relevant subsections.

Reform	Lever	Static revenue (£m)
Raise threshold to £100,000	Location	−374
Graduated taper [£85k, £105k]	Shape (phase-in)	−335
Reduced rate 10% [£85k, £105k]	Rate (step)	−357
Reduced rate 15% [£85k, £105k]	Rate (step)	−178

Table 3: Efficiency ranking of the reform menu. The dominated region (T^* , $T^* + a$), $a = T^* \tau / (1 - \tau)$ (Section 6.3), is a pure misallocation zone of width £21,250 under the current £85,000/20% notch. “Distortion removed” is the reduction in the width of this zone; static revenue is from Table 2. This is a robust, analytic deadweight/misallocation comparison based on the dominated region, *not* a full welfare integral (which would require value added and input reclaim; see Section 9). Reforms are ordered from least to most distortion removed.

Reform	Lever	Static rev. (£m)	Dominated region	Distortion removed
Raise threshold to £100,000	Location	−374	£21,250 (relocated)	$\approx$ £0
Reduced rate 15%	Rate	−178	£15,000	£6,250
Reduced rate 10%	Rate	−357	£9,444	£11,806
Graduated taper	Shape	−335	£0 (no notch)	£21,250 (whole zone)

Table 4: Summary of headline estimates. This table consolidates results reported and defined in the body sections cross-referenced below; it is a reference summary, not a new analysis. All quantities refer to the £85,000 data-year baseline unless a reform threshold is stated. Static revenue figures are 2025–26.

Object	Symbol / definition	Value	Source
<i>Reduced-form bunching (£85k data year)</i>			
Bunching ratio (geometry, <i>not</i> an elasticity)	$b = \Delta y^* / y^*$	0.061	§5
Aggregate displaced share	Π	0.220	§7
<i>Behavioural response (headline object)</i>			
Local turnover elasticity (median)	$\tilde{\epsilon}$	≈ 0.17	§7
Near-threshold firms repositioning	—	$\approx 15\%$	§7
<i>Policy costing and efficiency</i>			
Static cost, £85k to £90k anchor	—	$\approx -£175\text{m}$	§8
Static sweep, £90k baseline	per £5k step	$\sim £150\text{--}220\text{m}$	Table 1
Graduated-taper reform, revenue (static)	—	$-£335\text{m}$	Table 2
Reduced-rate band, revenue static (10% / 15%)	—	$-£357\text{m} / -£178\text{m}$	Table 2
Dominated region (misallocation zone)	$a = T^* \tau / (1 - \tau)$	$£21,250$	§6.3
Most efficient reform	taper removes whole zone	—	Table 3

roughly 10% over the band, so two differently-shaped schedules impose nearly the same average burden. I label the efficiency comparison honestly: it is a deadweight/misallocation ranking based on the dominated-region distortion, which is robust and analytic, not a full welfare integral. Converting these distortion measures into welfare or MVPF terms requires modelling value added and input reclaim and is left to future work (Section 9).

9 Conclusion

This paper contributes an open, reproducible microsimulation for costing UK VAT registration-threshold reforms, and uses it to price and *rank* a menu of reforms by efficiency. The framework—a synthetic firm dataset calibrated to HMRC VAT aggregates, a reduced-form bunching estimator, and a calibrated Kleven and Waseem (2013) notch model used to re-solve firm behaviour under alternative schedules—is released in full for replication. Its value is as usable policy infrastructure rather than as a structural identification result: the recovered cost parameters are a calibration input for re-solving reforms, not claimed structural primitives, and papers such as Garicano et al. (2016) and Gourio and Roys (2014), which do identify such primitives at a size threshold, are the precedent I differ from precisely by not making that claim. The analysis is anchored to the 2023–24 data year, in which the statutory threshold was £85,000. Table 4 collects the headline objects.

Findings. First, *the distortion is modest and localised*. Firms bunch below the notch, but the headline behavioural object—the local turnover elasticity (median ≈ 0.17)—is small, broadly comparable to the UK VAT-base elasticity of Liu et al. (2021), though the two are distinct objects, so the comparison is an order-of-magnitude sanity check. Only a minority of near-threshold firms reposition, and within a narrow band.

Second, *threshold-location moves are fiscally small and just relocate the notch*. Moving the threshold is approximately linear in revenue (Table 1) and adequately costed by a static calculation; as a pipeline sanity check, my static costing of the real £85,000 to £90,000 reform agrees with HMRC’s published figures to within £1m in 2025–26. But raising the threshold does not remove the notch’s distortion—it transports the same £21,250 dominated region to the new cutoff at a comparable fiscal cost.

Third—and centrally—*schedule reforms remove the distortion, and the taper is the efficiency winner*. The notch creates a dominated region of width $a = T^* \tau / (1 - \tau) = £21,250$, a pure misallocation zone in which no firm optimally locates. Ranking the menu on how much of this zone each reform removes per pound of static cost (Table 3), a location move removes essentially none; a banded reduced rate shrinks it (to £9,444 at 10%); and the graduated taper eliminates it entirely, at a static cost (–£335m) below that of the £85,000 to £100,000 threshold move (–£374m). Schedule reforms thus dominate location moves on efficiency, and the taper—a continuous phase-in that leaves no discrete wedge and no secondary notch—is the most efficient option in the menu. This is a robust, analytic deadweight/misallocation comparison based on the dominated region; the static revenue figures are robust to the smoothing parameter, while the behavioural re-optimised magnitudes are sensitive to it and are reported only as directional. Because a smooth or banded schedule leaves no single point of excess mass, these reforms are exactly what the calibrated model is needed for.

Limitations. The model is a *turnover-tax-notch approximation*, and this is its first-order limitation. It treats VAT as a cost on turnover, whereas real VAT is levied on value added with input reclaim, and roughly half of below-threshold firms register *voluntarily* because reclaim outweighs the output liability—the central fact in Liu et al. (2021). The effective notch is therefore smaller (and, for net-reclaim firms, can reverse sign) than my flat treatment implies. The microdata are *synthetic*—ONS business-structure counts calibrated to HMRC aggregates—so agreement with HMRC totals is a consistency check, not external validation. The analysis is *partial-equilibrium*, abstracting from VAT pass-through into prices (Benedek et al., 2015; Bellon et al., 2026). And while the dominated-region efficiency ranking is robust and analytic, the *behavioural* revenue and distributional magnitudes are parameter-sensitive (to the smoothing parameter k and the re-optimisation window), which is why I headline the static costs and the analytic distortion measures rather than the re-optimised numbers.

Future work. The most direct extension is a full *welfare/MVPPF analysis* that models value added and input reclaim, converting the dominated-region distortion measures into the cost–benefit terms policy needs, and the design of *revenue-neutral* variants of the taper and reduced rate that neutralise the notch at no net fiscal cost. Modelling

voluntary registration and the input-VAT margin explicitly would let the framework represent the full UK VAT population rather than the consumer-facing subpopulation it currently best describes. Allowing sector heterogeneity and linking the microdata to [PolicyEngine \(2025\)](#) would refine both the bunching estimates and the counterfactuals. Above all, *validation against administrative firm-level data* would replace the present internal-consistency checks with genuine out-of-sample tests.

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A Appendix

This appendix collects supporting material in three parts: data construction and calibration robustness (Appendix A.1); robustness of the structural model to the smooth-notch approximation (Appendix A.2); and statistical inference for the reduced-form and structural estimates (Appendix A.3).

A.1 Data construction detail

Section 3 describes the two-stage synthetic-population pipeline (`analysis/gen_full_param.py`, parameterised by `VAT_THRESHOLD`) and its calibration to HMRC and ONS aggregates. This appendix records the implementation detail deferred from that account.

Each of the approximately 2.94 million firm rows carries the fields `annual_turnover_k`, `annual_input_k`, `weight`, and `vat_registered`; calibration weights are chosen so the weighted totals reproduce the official targets while the population sums to about 2.0 million firms. Because the population is calibrated *to* the HMRC aggregates, agreement between my weighted outputs and those aggregates is a check of *internal consistency*, not external validation: any target used in calibration is reproduced by construction. I therefore read the $\sim 87\%$ calibration accuracy, and the close match of the static $\pounds 85,000 \rightarrow \pounds 90,000$ costing to HMRC’s published figures, as consistency checks on the pipeline rather than out-of-sample tests of firm behaviour.

Counterfactual exclusion window and polynomial degree. The no-VAT counterfactual density of Section 6 is fitted by polynomial regression on the observed density outside a manipulation window around the threshold. The baseline excludes $[T^* - 15k, T^* + 15k]$ and fits a cubic. Widening the window guards against contamination of the counterfactual by the behavioural response but reduces the data available for the fit; raising the degree improves in-sample fit but risks over-fitting the tails. The sensitivity of the excess-mass and bunching-ratio estimates to window width and degree, with associated uncertainty, is reported in Appendix A.3.

Subsample for dynamic plots. The dynamic re-optimisation figures (Figure 8) are generated on a 20% random subsample of the firm rows, to keep the firm-by-firm FOC inversion and re-solution tractable, while the static revenue sweep and the calibration use the full dataset. The 20% subsample reproduces the full-dataset density step and bunching mass to within sampling error, so subsampling affects only the smoothness of the plotted curves, not the substantive conclusions.

A.2 The sigmoid as a smoothed-notch approximation

The calibrated model of Section 6 treats the registration threshold as a true notch and characterises its analytic objects—the dominated region and the marginal buncher—exactly. For the forward solution at the level of two million firms, I operationalise the notch with a smooth (logistic) approximation to the tax schedule. This appendix sets out

that approximation and establishes when it is faithful: the smooth schedule reproduces the notch—including its dominated region—when its steepness k is calibrated to the Kleven–Waseem width, and degenerates towards a smoothed *kink* only when k is mis-set. I calibrate k accordingly and report all exact analytic objects from the true notch, so the smoothing affects only the numerics of the forward solution, not the paper’s quantitative claims.

The smoothing device

The exact notch makes the effective average rate a step function of turnover, non-differentiable at T^* . To obtain a smooth schedule, I replace the step by a logistic effective-rate schedule,

$$\tau(y_i) = \tau_{\max} \frac{1}{1 + e^{-k(y_i - T^*)}}, \quad \frac{\partial \tau(y_i)}{\partial y_i} = k \tau(y_i) \left(1 - \frac{\tau(y_i)}{\tau_{\max}}\right), \quad (8)$$

where $k > 0$ governs the steepness: as $k \rightarrow \infty$ the sigmoid converges pointwise to the exact step at T^* , and its derivative peaks at T^* and decays away from it. The corresponding optimality condition equates marginal after-tax revenue $(1 - \tau(y_i)) - y_i \partial \tau / \partial y_i$ to marginal cost. I do not solve this condition as a root-find: with $\alpha \approx 0.989$ the near-constant-returns specification makes a continuous interior optimum ill-conditioned, so the operational solver maximises after-tax profit directly over a bounded grid (described below), which does not rely on the optimality condition having a well-defined interior root.

Fidelity of the smooth approximation

Equation (8) smooths the *average* rate $\tau(y_i)$ applied to the firm’s entire turnover, so after-tax revenue is $R(y_i) = (1 - \tau(y_i))y_i$. Whether this reproduces the institutional notch depends on the steepness k . Under the true notch, after-tax revenue drops discretely by $\tau_{\max} T^* = \text{£}17,000$ at the threshold; under the smooth schedule the drop is spread over a transition whose width falls as k rises, and whether $R(y_i)$ decreases over a neighbourhood of T^* —the structural analogue of the output-dominated region—holds for steep k and fails for shallow k , where the schedule collapses to a smoothed kink that merely perturbs the marginal rate locally.

The consequences for the dominated region are stark. Table A.1 reports the width of the locally output-dominated region implied by the sigmoid (the band where marginal after-tax revenue $(1 - \tau) - y\tau'$ turns negative) against the exact Kleven–Waseem width $a = T^* \tau_{\max} / (1 - \tau_{\max}) = \text{£}21,250$ of (4). Only at a steepness close to $k = 0.001$ does the implied band approach the exact width; at the larger $k = 0.01$ used in earlier code the sigmoid captures barely a tenth of it, and for very small k the schedule is monotone and exhibits *no* dominated region at all—a pure kink. The full k grid and its reconciliation with the empirical excess span are reported in Appendix A.3.

Two points reinforce the message. First, the sigmoid’s dominated-region width is almost entirely a function of k , which was not calibrated to any data moment in earlier

Table A.1: Sigmoid steepness k and the implied output-dominated region, compared to the exact Kleven–Waseem notch width $a = T^* \tau_{\max} / (1 - \tau_{\max}) = £21,250$ at $T^* = £85,000$, $\tau_{\max} = 0.20$. The true notch is k -independent.

Schedule	Dominated-region width	Character
Sigmoid, $k = 0.001$	$\approx £19.7\text{k}$	notch-like
Sigmoid, $k = 0.01$	$\approx £2.5\text{k}$	narrow, near-kink
Sigmoid, $k = 0.0001$	none (monotone R)	pure kink
True notch (this paper)	$£21,250$ (exact)	notch

drafts and took inconsistent values across paper and code ($k = 0.01$ versus $k = 0.001$); any bunching prediction read off the sigmoid inherits this arbitrariness. Second, for the questions this paper asks—the size of the bunching band, the location of the marginal buncher, and the revenue consequences of moving the notch—these discrepancies are first-order, which is why the main text models the notch exactly.

Recommended use and reconciliation

I use the smooth schedule of (8) as the effective-rate schedule in the forward firm-by-firm re-optimisation, but the firm’s optimum is located by *bounded grid maximisation of after-tax profit*, not by an interior root-find. For each firm I evaluate after-tax profit on a fine turnover grid over a bounded neighbourhood ($\pm£20\text{k}$ on a $£200$ step) of its observed turnover and select the profit-maximising point. I adopt direct profit maximisation precisely because the near-constant-returns specification ($\alpha \approx 0.989$) makes the interior optimum degenerate, so the global maximiser is well-defined while an interior root is not. The $\pm£20\text{k}$ window is itself a modelling assumption, motivated by the documented localisation of the realised response within roughly $£20,000$ of the threshold (Section 8.1); I note that this choice mechanically bounds how far firms can move, so the “localised, partial” response partly reflects the window rather than being a free output of the exercise.

I adopt two further safeguards throughout. (i) I calibrate k to the exact dominated-region width $a = T^* \tau_{\max} / (1 - \tau_{\max})$ ($k \approx 0.001$, which reproduces the band to within about 7%; Table A.1) rather than asserting it, and use a single value consistently. (ii) I report every exact analytic object—the $\sim £21\text{k}$ dominated region and the marginal buncher of Section 6.4—from the true notch of Section 6, not from the smoothing. Consistent with this posture, I headline the taper’s robust revenue effect and its analytic efficiency mechanism (the removal of the dominated region and the discrete wedge); the *simulated* change in the near-threshold turnover distribution depends on both k and the re-optimisation window and is not reported as a magnitude (Section 8.2).

A.3 Inference for the bunching estimator

This appendix documents the inference procedure for the reduced-form bunching estimates of Section 7: the mass-conservation constraint that disciplines the counterfactual density, the bootstrap that delivers standard errors, and the sensitivity of the estimates to the polynomial degree, the exclusion window, the sigmoid steepness k , and the effective wedge τ_e . All numbers reported here and in the calibration section are produced by the script `analysis/bunching_inference.py`, run on the £85,000 (2023–24) weighted synthetic firm data; nothing is hardcoded as a result.

Mass-conservation constraint

The canonical bunching estimator of Saez (2010), Kleven and Waseem (2013) and Chetty et al. (2011) imposes that the counterfactual density conserves mass: every firm removed from above the threshold reappears as excess mass below it. An unconstrained fit by plain polynomial OLS on bins outside a hand-set window computes the excess mass E (below T^*) and the missing mass Δ_R (above T^*) *independently*, with no guarantee that $E = \Delta_R$ and no endogenous location of the marginal buncher.

I impose the constraint in two ways. First, the counterfactual polynomial in $(y - T^*)$ is fit by an iterated weighted least-squares procedure in which the density of the bins above the excluded region is rescaled until the fitted counterfactual integrates to the *observed* total mass over the estimation range [£20k, £140k]. This forces the bunching mass to be reabsorbed into the counterfactual rather than discarded. The regressor is normalised to $[-1, 1]$ for numerical stability of the high-degree Vandermonde basis. Second, the upper edge of the excluded (manipulation) region—the *marginal buncher* y_R —is located *endogenously* by integrating the missing mass upward from T^* until cumulative missing mass above equals cumulative excess mass below,

$$\int_{T^*}^{y_R} [f^{\text{cf}}(y) - f^{\text{obs}}(y)]_+ dy = \int_{T^*-W}^{T^*} [f^{\text{obs}}(y) - f^{\text{cf}}(y)]_+ dy = E,$$

rather than being fixed at an arbitrary window edge. The excess turnover span of the marginal buncher, $\Delta y^* = y_R - T^*$, is then used by the notch-width elasticity (Method B).

Imposing the constraint matters because the unconstrained fit grossly violates adding-up: it delivers $E = 8,179$ but $\Delta_R = 17,792$, whereas the constrained estimator returns $E = 9,160$ and $\Delta_R = 9,938$, which agree to within sampling noise. The accompanying movement in the headline estimates (b from 0.054 to 0.061, σ from 4.33 to 5.10) is reported in Section 7.

Bootstrap standard errors

I compute standard errors by a firm-level bootstrap. In each of $B = 200$ replications I draw n firms with replacement from the estimation sample, with draw probabilities proportional to the survey weight, re-bin the resampled firms into £1,000 bins, and re-run

Table A.2: Bootstrap inference, £85,000 data ($B = 200$ firm resamples).

Statistic	Point	Std. error	95% CI
Bunching ratio b	0.061	0.005	[0.053, 0.071]
Excess mass E	9,160	672	[8,325, 10,931]
Substitution elasticity σ	5.10	0.12	[4.86, 5.35]
Displaced share Π	0.220	0.005	[0.211, 0.230]
Notch-width elasticity (median)	0.458	0.027	[0.429, 0.529]
Marginal-buncher elasticity	0.635	0.077	[0.514, 0.806]

Table A.3: Degree $\times$ window sensitivity (point estimates), £85,000 data. Cells with $b < 0$ leave σ undefined.

Degree	Window	b	E	σ	Notch-width (median)
6	£10k	0.079	7,719	5.69	0.42
6	£15k	0.046	6,900	5.24	0.45
6	£20k	0.017	4,101	4.44	0.38
7	£10k	0.095	9,073	5.32	0.47
7	£15k	0.061	9,160	5.10	0.46
7	£20k	0.036	7,785	4.89	0.45

the entire estimator—counterfactual fit, marginal-buncher location, and all downstream statistics—on the replicate. The standard error of each statistic is the standard deviation of its bootstrap distribution, and the 95% confidence interval is the 2.5th–97.5th percentile interval. Resampling firms (rather than residuals) propagates both sampling variation and the variation induced by re-binning. Table A.2 reports the results.

Sensitivity to polynomial degree and exclusion window

Table A.3 maps the point estimates across a grid of polynomial degree $\in \{5, 6, 7, 8\}$ and symmetric exclusion window $\in \{\text{£10k}, \text{£15k}, \text{£20k}, \text{£25k}\}$. The estimates are stable across the central cells (degree 6–7, window £10–£20k): the substitution elasticity stays in roughly 4.4–5.7 and the excess mass in 4,000–9,000. The corner cells are less reliable: at very wide windows (£25k) the excluded region begins to swallow genuine curvature in the density, the fitted counterfactual crosses the observed density, and b can turn slightly negative (so σ is undefined); at the lowest degree (5) the inflexible trend over-fits the bunching mass into the counterfactual and inflates b . I therefore report the degree-7, £15k-window specification as the baseline, with the bootstrap CIs above spanning the variation across the stable interior of the grid.

Sensitivity to the sigmoid steepness

The notch-width elasticity uses the effective wedge $\tau_{\text{eff}} = \tau_{\text{max}}/2$, which holds at the threshold for *any* sigmoid steepness k , so the data-side elasticity is k -invariant. The steepness k instead governs the *structural* model: it sets the width of the output-dominated (“forbidden”) band of the smoothed schedule, defined as the turnover region in which marginal net revenue $(1 - \tau(y)) - y\tau'(y)$ is negative. My code computes this width directly and compares it to the exact Kleven and Waseem (2013) dominated region for a 20% notch at £85,000, $T^*\tau/(1 - \tau) = £21.25\text{k}$. The width is strongly k -dependent: $k = 0.0001$ leaves essentially the entire local band dominated (a near-kink), $k = 0.001$ gives £19.7k (closest to the exact £21.25k), $k = 0.002$ gives £10.6k, $k = 0.005$ gives £4.6k, and $k = 0.01$ gives only £2.5k; the empirical excess span of the marginal buncher is $\Delta y^* = £5.4\text{k}$. This is the basis for the choice of $k = 0.001$ in Section 7: its implied dominated region best matches both the exact notch region and the order of magnitude of the empirical span, whereas $k = 0.01$ understates the band by roughly an order of magnitude. I report results at $k = 0.001$ and treat the spread above as a calibration sensitivity rather than a point identification of k .

Sensitivity to the effective wedge

The CES substitution elasticity is recovered as $\sigma = \ln(\text{RR})/\ln(1 + \tau_e)$, where RR is the ratio of counterfactual-to-observed above/below mass ratios. Because the wedge enters only through $\ln(1 + \tau_e)$, σ scales roughly as $1/\ln(1 + \tau_e)$ while the displaced share $\Pi = 1 - (1 + \tau_e)^{-\sigma}$ is essentially invariant (it is pinned by RR). On my data, $\tau_e = 0.025$ gives $\sigma = 10.1$, $\tau_e = 0.05$ gives $\sigma = 5.10$, $\tau_e = 0.075$ gives $\sigma = 3.44$, and $\tau_e = 0.10$ gives $\sigma = 2.61$, while $\Pi = 0.22$ throughout. The magnitude of σ is therefore conditional on τ_e : I adopt $\tau_e = 0.05$ as a baseline effective registration wedge and report Π , which is robust to it, as the policy-relevant displacement object.